Agisoft Metashape

Processing Report

08 September 2026



Survey Data

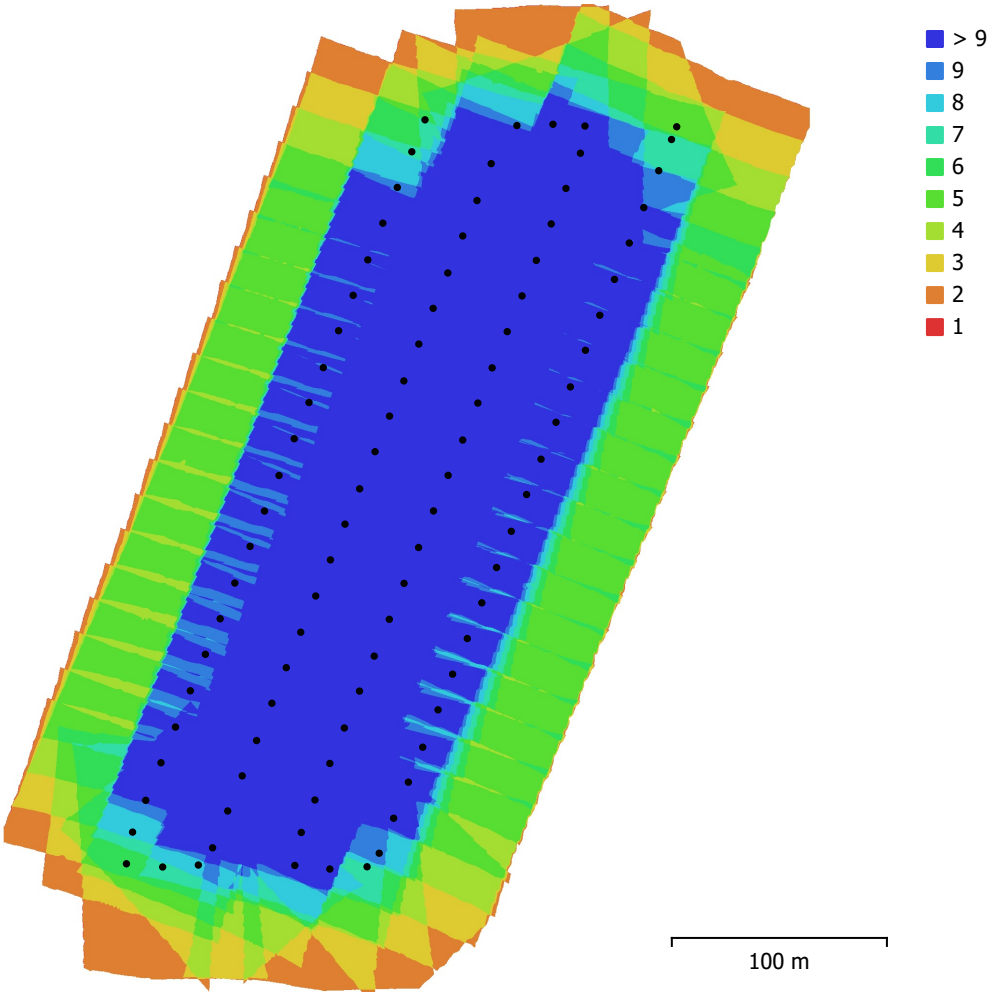


Fig. 1. Camera locations and image overlap.

Number of images:	92	Camera stations:	92
Flying altitude:	84.2 m	Tie points:	82,580
Ground resolution:	2.87 cm/pix	Projections:	221,975
Coverage area:	0.103 km²	Reprojection error:	0.559 pix

Camera Model	Resolution	Focal Length	Pixel Size	Precalibrated
ZH20T (4.5mm)	4056 x 3040	4.5 mm	1.6 x 1.6 µm	No

Table 1. Cameras.

Camera Calibration

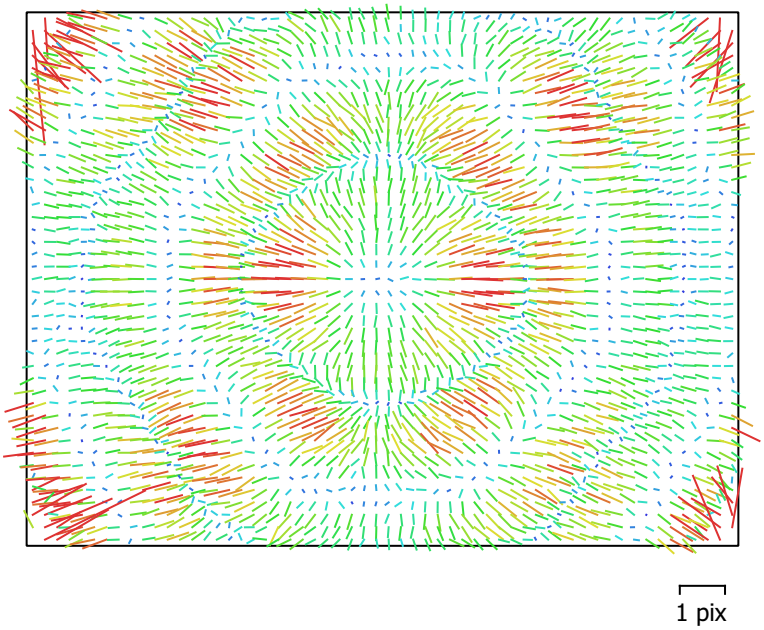


Fig. 2. Image residuals for ZH20T (4.5mm).

ZH20T (4.5mm)

92 images

Type	Resolution	Focal Length	Pixel Size
Frame	**4056 x 3040**	**4.5 mm**	**1.6 x 1.6 µm**

	Value	Error	Cx	Cy	B1	B2	K1	K2	K3	K4	P1	P2
F	**2811.66**											
Cx	**9.08688**	0.15	1.00	0.08	0.04	-0.09	0.06	0.02	-0.03	0.03	-0.56	-0.02
Cy	**13.9963**	0.12		1.00	0.08	0.02	-0.00	0.02	-0.02	0.02	-0.07	-0.23
B1	**-4.90819**	0.027			1.00	0.05	-0.06	-0.00	-0.00	0.01	-0.11	-0.09
B2	**-0.528766**	0.027				1.00	-0.05	0.00	-0.00	0.01	0.04	-0.13
K1	**0.263418**	0.00036					1.00	-0.75	0.71	-0.70	-0.02	0.26
K2	**-0.857658**	0.0012						1.00	-0.99	0.96	0.02	-0.06
K3	**1.05995**	0.0023							1.00	-0.99	-0.02	0.05
K4	**-0.528096**	0.0016								1.00	0.02	-0.06
P1	**5.33373e-05**	7.9e-06									1.00	-0.01
P2	**-0.000701888**	8.3e-06										1.00

Table 2. Calibration coefficients and correlation matrix.

Camera Locations

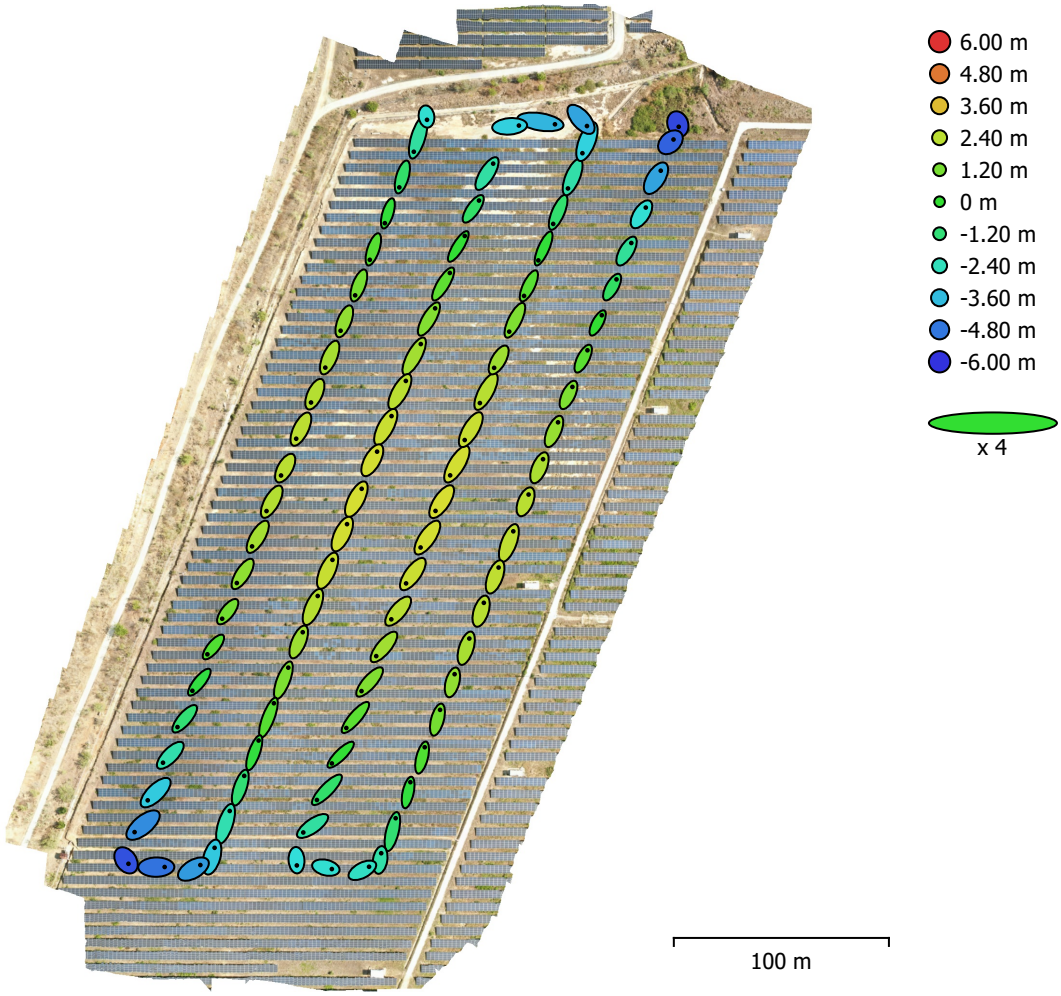


Fig. 3. Camera locations and error estimates.
Z error is represented by ellipse color. X,Y errors are represented by ellipse shape.
Estimated camera locations are marked with a black dot.

X error (m)	Y error (m)	Z error (m)	XY error (m)	Total error (m)
1.30436	2.15397	2.37452	2.51813	3.46112

Table 3. Average camera location error.
X - Longitude, Y - Latitude, Z - Altitude.

Digital Elevation Model

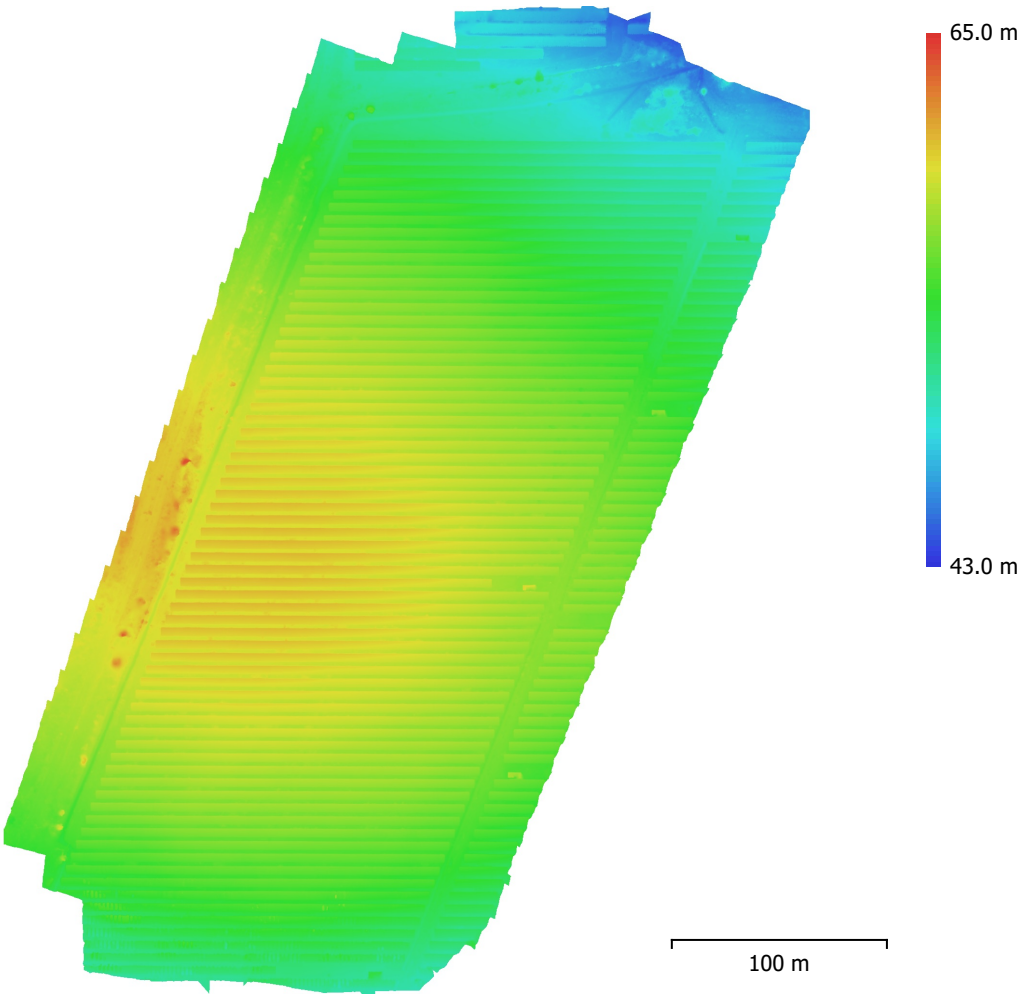


Fig. 4. Reconstructed digital elevation model.

Resolution:	5.99 cm/pix
Point density:	279 points/m²

Processing Parameters

General	
Images	92
Aligned images	92
Coordinate system	WGS 84 (EPSG::4326)
Rotation angles	Yaw, Pitch, Roll
Tie Points	
Points	82,580 of 137,023
RMS reprojection error	0.229615 (0.559013 pix)
Max reprojection error	1.51623 (4.69593 pix)
Mean key point size	2.47027 pix
Point colors	3 bands, uint8
Key points	302.52 MB
Average tie point multiplicity	2.91812
Alignment parameters	
Accuracy	High
Generic preselection	Yes
Reference preselection	Source
Key point limit	40,000
Key point limit per Mpx	1,000
Tie point limit	4,000
Exclude stationary tie points	Yes
Guided image matching	No
Adaptive camera model fitting	Yes
Matching time	2 minutes 57 seconds
Matching memory usage	455.00 MB
Alignment time	1 minutes 9 seconds
Alignment memory usage	278.34 MB
Optimization parameters	
Parameters	f, cx, cy, k1, k2, p1, p2
Adaptive camera model fitting	Yes
Exclude corners	No
Optimization time	2 seconds
Date created	2026:08:12 17:09:59
Software version	2.3.1.22580
File size	312.45 MB
Depth Maps	
Count	92
Depth maps generation parameters	
Quality	High
Filtering mode	Aggressive
Max neighbors	16
Processing time	3 minutes 33 seconds
Memory usage	1.82 GB
Date created	2026:08:12 17:27:15
Software version	2.3.1.22580
File size	455.71 MB
Point Cloud	
Points	37,577,881
Coordinate precision	1.44 cm
Point attributes	

Color	3 bands, uint8
Normal	
Point classes	
Created (never classified)	37,577,881
Depth maps generation parameters	
Quality	High
Filtering mode	Aggressive
Max neighbors	16
Processing time	3 minutes 33 seconds
Memory usage	1.82 GB
Point cloud generation parameters	
Source data	Depth maps
Processing time	6 minutes 49 seconds
Memory usage	2.34 GB
Date created	2026:08:12 17:34:07
Software version	2.3.1.22580
File size	501.33 MB
DEM	
Size	6,252 x 7,662
Resolution	5.99 cm/pix
Coordinate system	WGS 84 (EPSG::4326)
Depth maps generation parameters	
Quality	High
Filtering mode	Aggressive
Max neighbors	16
Processing time	3 minutes 33 seconds
Memory usage	1.82 GB
Reconstruction parameters	
Source data	Depth maps
Interpolation	Enabled
Processing time	21 minutes 2 seconds
Memory usage	4.08 GB
Date created	2026:08:12 19:09:25
Software version	2.3.1.22580
File size	123.22 MB
Orthomosaic	
Size	12,504 x 15,324
Resolution	2.99 cm/pix
Coordinate system	WGS 84 (EPSG::4326)
Colors	3 bands, uint8
Orthophotos	597.59 MB
Reconstruction parameters	
Blending mode	Mosaic
Surface	DEM
Enable hole filling	Yes
Enable ghosting filter	No
Processing time	2 minutes 13 seconds
Memory usage	1.11 GB
Date created	2026:08:12 22:18:01
Software version	2.3.1.22580
File size	799.38 MB
System	
Software name	Agisoft Metashape Professional
Software version	2.3.0 build 21954
OS	Windows 64 bit
RAM	15.78 GB

CPU	11th Gen Intel(R) Core(TM) i5-11400H @ 2.70GHz
GPU(s)	NVIDIA GeForce RTX 3050 Laptop GPU